Manual Identification of Malaysia's Statutory PIT-Rate Shocks

AI-assisted narrative pass over the C1 / C0 deployment evidence

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1 Purpose

This note documents a focused, **AI-assisted manual identification** of Malaysia's *statutory personal/individual income-tax (PIT) rate changes*, run directly over the evidence our deployment pipeline already surfaced. It is the PIT counterpart to `notebooks/cit_identification.qmd`, and it follows the same spirit of the narrative-identification literature (Romer and Romer 2010): close reading of source documents, with the LLM pipeline as a retrieval-and-extraction aide, not as the adjudicator.

The method mirrors our `/manual-analysis` protocol. The pipeline (C1 measure identification, C0 act aggregation) pre-screens candidate measures; a human reads the surfaced evidence and makes the final call on *what counts as an event*, *how scattered surface forms consolidate*, and *whether each change is exogenous*. Claude pattern-matches against the source text; the researcher confers meaning.

PIT differs from CIT in one structural respect: it is a **progressive schedule, not a single rate**. We track the **top marginal resident rate** as the headline series and record bracket/threshold changes as detail. Two of the nine identified acts move brackets without moving the top rate, and are flagged as such.

> ! Important
>
> The exogenous/endogenous column in Table 3 is a **preliminary suggestion with its supporting quote, pending expert adjudication**. It is not a research finding. The whole point of this pass is that a human adjudicates these calls; see the open questions in Section 6.

2 What C1 surfaced, and what it missed

A keyword scan of the full C1 pool returns **30 distinct PIT measures** spanning 1999–2023, in both English and Malay. A broader semantic sweep over all 1036 measures (reading every income-tax surface form, not just keyword hits) confirms the top-marginal-rate path runs **30 → 29 → 28 → 27 → 26 → 25**, then reverses upward to **28** (2016) and **30** (2020).

The sweep also found two **genuine recall misses**: headline events that sit in the corpus with explicit rates but that C1 did not surface as measures. Both were recovered by reading the source text directly.

Table 1: **Recall scorecard**. How each headline event fared through C1 measure identification and C0 act aggregation. 'Recovered manually' marks events present in the corpus that C1 did not surface as a measure.

Stage	Outcome
Keyword scan (C1 pool, n=1,036)	30 hits over the PIT regex spanning 1999-2023; 2 false positives excluded (a 2004 corporate 20%-band item; a SOCSO taxi-driver relief item).
Semantic sweep (all income-tax surface forms)	Confirmed the top-marginal-rate path 30->29->28->27->26->25 then back up to 28 (YA2016) and 30 (YA2020); relief/rebate/threshold tracks recorded as magnitude detail, not headline rows.
C1 measure identification	Surfaced P1, P2, P4, P5, P7, P8, P9; MISSED P3 (28->27, Budget 2009) and P6 (25->28 high-earner hike, Budget 2016).
Recall miss - Budget 2009 (28->27, YA2009)	Recovered from Budget Speech 2009 ch2 (BM: 'kadar cukai individu maksimum diturunkan dari 28 peratus kepada 27 peratus, mulai tahun taksiran 2009'); retrospectively confirmed in Budget Speech 2010 ch6.
Recall miss - Budget 2016 high-earner hike (25->28, YA2016)	Recovered from Budget Speech 2016 ch3/9 + Appendix 1 ('chargeable income exceeding RM1,000,000 be increased by 3 percentage points … 25% … 28%').
C0 act aggregation	Not used as the event layer (same fragmentation finding as CIT: near-all singletons; EN/BM never bridged).

The retrieval lesson is the same one the CIT pass surfaced: a statutory rate change is maximally relevant *because it is a rate change*, however briefly the documents discuss it, so an instrument-targeted recall pass (plus human reading) is needed rather than trust in C0's clustering. Both PIT recall misses are top-of-schedule moves (one cut, one hike) that the broad measure-identification pass passed over.

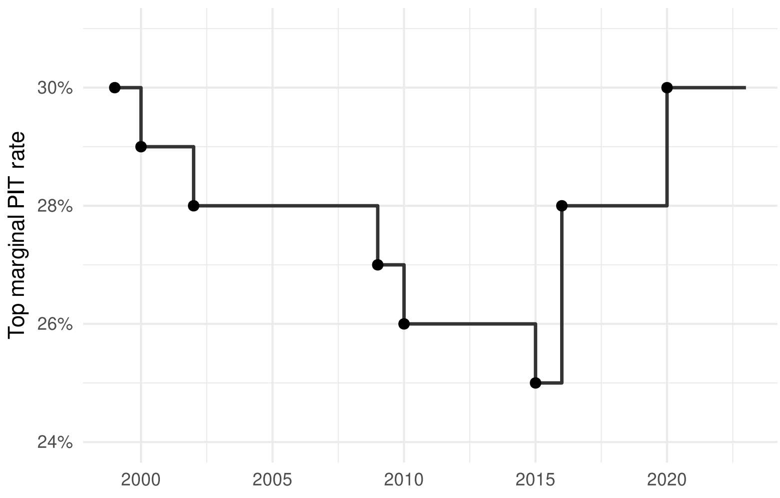
3 The headline statutory top PIT rate

Reading the consolidated evidence yields a clean headline path for the **top marginal resident rate**: **30 → 29 → 28 → 27 → 26 → 25 → 28 → 30**. Table 2 gives the rate-moving events; Figure 1 traces the rate. Two further acts (a 2018 middle-band cut and the 2023 M40/T20 restructure) change the schedule without moving the top rate and are noted below the figure.

Table 2: **Headline top-PIT-rate events**. Identified statutory changes in the top marginal resident income-tax rate, with the budget that announced each and the year of assessment it took effect. Acts that move only lower brackets (P7, P9) are excluded here and discussed in Section 5.

Event	From->To	Dpp	Direction	Effective (YA)	Announced
MY-PIT-01	30% -> 29%	−1	Cut	2000	1999
MY-PIT-02	29% -> 28%	−1	Cut	2002	2001
MY-PIT-03	28% -> 27%	−1	Cut	2009	2008
MY-PIT-04	27% -> 26%	−1	Cut	2010	2009
MY-PIT-05	26% -> 25%	−1	Cut	2015	2013
MY-PIT-06	25% -> 28%	+3	Hike	2016	2015
MY-PIT-08	28% -> 30%	+2	Hike	2020	2019
MY-PIT-09	30% -> 30%	+0	Neutral	2023	2023

Figure 1: **Headline top marginal PIT rate, Malaysia 1999-2023**. Step path of the top resident individual income-tax rate from the identified events. The 2016 and 2020 steps are high-earner band introductions (>RM1m, >RM2m) that raised the statutory top rate; the 2018 middle-band cut and the 2023 M40/T20 restructure (top rate unchanged) are not part of this path.



The path is not monotone: a long sequence of competitiveness-driven cuts (2000–2015) is followed by two high-earner hikes (2016, 2020) that reintroduced rates above the prior 25% floor for the very top of the distribution. This non-monotonicity, and the mix of cut and hike framings, is exactly what makes the exogeneity adjudication consequential rather than mechanical.

4 Motivation evidence and preliminary exogeneity

For each act, the corpus carries motivation language. Table 3 pairs each event with the most diagnostic quote and a **preliminary** exogenous/endogenous read. The cuts of 2009 and 2010 carry explicit competitiveness/brain-drain framing; the 2000 cut sits inside the post-crisis demand-recovery package; the high-earner hikes and the 2023 restructure carry progressivity framing amid fiscal pressure.

Table 3: **Motivation evidence, pending adjudication**. The stated rationale for each act and a preliminary exogenous/endogenous read. These are suggestions anchored in the quoted text, not settled classifications; the exogeneity calls are the object of expert adjudication (see Section 6).

Event	Motivation quote (source text)	Preliminary read
MY-PIT-01	one percentage point across-the-board reduction in personal income tax rates, higher tax relief and salary adjustments ... real aggregate domestic demand is expected to strength...	FALSE
MY-PIT-02	kadar cukai pendapatan individu ... itu menyamai kadar cukai pendapatan korporat (top individual rate aligned with corporate rate; competitiveness, Budget 2002)	ambiguous
MY-PIT-03	Sebagai salah satu langkah untuk menjadikan kadar cukai individu kompetitif dan menarik ... membendung brain drain ... kadar cukai individu maksimum diturunkan dari 28 peratus k...	TRUE
MY-PIT-04	To ensure that the individual income tax remains competitive ... reduced from 27% to 26% effective from the year of assessment 2010 (Budget Speech 2010)	TRUE
MY-PIT-05	REVIEW OF INDIVIDUAL INCOME TAX ... rates ... be reduced by 1 percentage point to 3 percentage points (Budget 2014; GST-offset package, effective YA2015)	ambiguous
MY-PIT-06	To enhance the progressivity of the individual income tax structure ... chargeable income exceeding RM1,000,000 be increased by 3 percentage points ... 25% to 28%. From year of ...	ambiguous
MY-PIT-07	I would like to announce an individual income tax rates reduction of two percentage points for chargeable income tax band between RM20,000 to RM70,000 (Budget 2018)	ambiguous
MY-PIT-08	new band for chargeable income in excess of RM2 million will be introduced and taxed at 30%, which is a 2 percentage point increase from the current 28% rate. This increase will...	ambiguous
MY-PIT-09	the income tax rates for resident individual will be reduced by 2% for income ranges from RM35,000 to RM100,000 ... the Government will raise tax rates for those with higher inc...	ambiguous

5 Bracket-only acts and non-rate tracks

Per the PIT scope, the top marginal rate is the headline series; sub-top changes are recorded as detail. Table 4 lists the two acts that change the schedule without moving the top rate, plus the relief/rebate/threshold tracks surfaced by C1 that are not rate shocks.

Table 4: **Bracket-only acts and non-rate tracks**. The 2018 middle-band cut and the 2023 M40/T20 restructure leave the top rate unchanged and are flagged as bracket changes. Reliefs, rebates and threshold moves are tracked as magnitude detail, not headline rows.

Track	Identified events	Note
Middle-band cut (P7)	Budget 2018: 2pp cut on RM20k-70k bands (5->3, 10->8, 16->14%)	Top marginal rate unchanged at 28%; kept as a row, rate fields NA
M40/T20 restructure (P9)	Budget 2023: M40 −2pp (RM35k-100k); T20 +0.5-2pp (RM100k-1m)	Top rate unchanged at 30%; direction Neutral, split in magnitude_note
Reliefs / rebates	2001/2002 rebates; personal relief raised RM8k->RM9k (2010); tuition/parental/SOCSO reliefs (2016)	Base-narrowing, not statutory-rate shocks
Threshold / structure	Chargeable-income band regrading (2002, 2015)	Recorded in magnitude_note for the parent rate event

6 Open adjudication questions

Four calls are genuinely the researcher's to make; they were adjudicated at the `/identify-pit` stamp (the frozen dataset in Section 7 encodes the decisions).

Resolved at freeze (2026-06-25)

1. **Mid-band-only acts kept as rows.** P7 (Budget 2018) and P9 (Budget 2023) are kept as one row per narratively-announced act, flagged in `magnitude_note` and `id_reasoning` as bracket changes that do not move the top marginal rate (P7 carries `NA` rate fields; P9 is `Neutral` with `delta_pp = 0`).
2. **P5 (YA2015 GST-offset cut) frozen as `ambiguous`**, its own PIT row (not merged with the GST act), mirroring the CIT GST-offset call (`MY-CIT-08`).
3. **P9 direction `Neutral`.** The mixed M40-cut / T20-hike restructure leaves the top rate unchanged; the split is preserved in `magnitude_note` as the object of adjudication.
4. **P1 top from/to `30 -> 29`.** Inferred from the documented "across-the-board 1pp" cut plus the 29% pre-2002 top rate; flagged in `id_reasoning`.

The narrative above is retained as the reasoning record behind these calls.

i 1. Exogenous vs endogenous across the cut sequence

The 2009 and 2010 cuts carry explicit competitiveness/brain-drain framing (preliminary exogenous); the 2000 cut sits in the post-crisis recovery package (preliminary endogenous); the 2002 cut blends competitiveness with a recovery budget (ambiguous). The same identifying separation that drives the CIT pass applies here.

i 2. The high-earner hikes

The 2016 (>RM1m, 25->28) and 2020 (>RM2m, 28->30) hikes are framed as progressivity but landed amid fiscal pressure (low oil/commodity revenue). Progressivity (long-run, exogenous) or deficit-driven (still exogenous, different reason)? Both are frozen `ambiguous`.

i 3. The 2023 restructure

A simultaneous M40 cut and T20 hike with the top rate unchanged: a single redistributive act, or two conceptual legs? Frozen as one `Neutral` row, pending adjudication.

7 Structured frozen dataset

The narrative identification above is the human-curated content; the machine-readable form is frozen by the `/identify-pit` skill to `data/validated/MY_PIT_shocks.qs`, following the shared contract in `docs/phase_1/tax_shock_schema.md` (one row per announced act × tax type, with a `member_chunks` manifest linking each shock to its corpus chunks and a mandatory recall scorecard). That frozen file — not this notebook's tribbles — is the reference input the deployment pipeline binds and feeds to C2a/C2b for motivation and exogeneity (`tar_read(tax_shocks)`).

Section 2 above is the recall scorecard for this pass. The frozen dataset and its per-shock member-chunk manifest render below once the skill has produced it.

Table 5: **Frozen PIT shock dataset**. Structured statutory PIT events as stamped by the `/identify-pit` skill (selected columns). The preliminary exogeneity read is pending expert adjudication and is carried alongside — never replaced by — C2b's downstream label.

shock_id	act_label	direction	rate_from	rate_to	Dpp	Effective	Preliminary exo.
MY-PIT-01	Across-the-board individual income tax cut 1pp (Budget 2000)	Cut	30	29	−1	2000	FALSE
MY-PIT-02	Top individual income tax rate reduction 29% to 28% (Budget 2002)	Cut	29	28	−1	2002	ambiguous
MY-PIT-03	Top individual income tax rate reduction 28% to 27% (Budget 2009)	Cut	28	27	−1	2009	TRUE
MY-PIT-04	Top individual income tax rate reduction 27% to 26% [illegible]	Cut	27	26	−1	2010	TRUE
MY-PIT-0[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	ambigu-

Table 6: **Member-chunk manifest**. Number of corpus chunks attributed to each shock, and how many were recovered (present in the corpus but not surfaced by C1). The recovered chunks drive the downstream C2a re-run.

shock_id	act_label	Member chunks	Recovered chunks
MY-PIT-01	Across-the-board individual income tax cut 1pp (Budget 2000)	2	0
MY-PIT-02	Top individual income tax rate reduction 29% to 28% (Budget 2002)	8	0
MY-PIT-03	Top individual income tax rate reduction 28% to 27% (Budget 2009)	2	1
MY-PIT-04	Top individual income tax rate reduction 27% to 26% (Budget 2010)	2	0
MY-PIT-05	Individual income tax restructure, top rate 26% to 25% (Budget 2014, GST-offset)	2	0
MY-PIT-06	High-earner individual income tax hike 25% to 28% above RM1m (Budget 2016)	2	2
MY-PIT-07	Middle-band individual income tax cut 2pp, RM20k-70k (Budget 2018)	4	0
MY-PIT-08	New 30% top band above RM2m, 28% to 30% (Budget 2020)	1	0
MY-PIT-09	Individual income tax restructure: M40 cut / T20 hike (Budget 2023)	4	1

References

Romer, Christina D., and David H. Romer. 2010. "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks." *American Economic Review* 100 (3): 763–801.